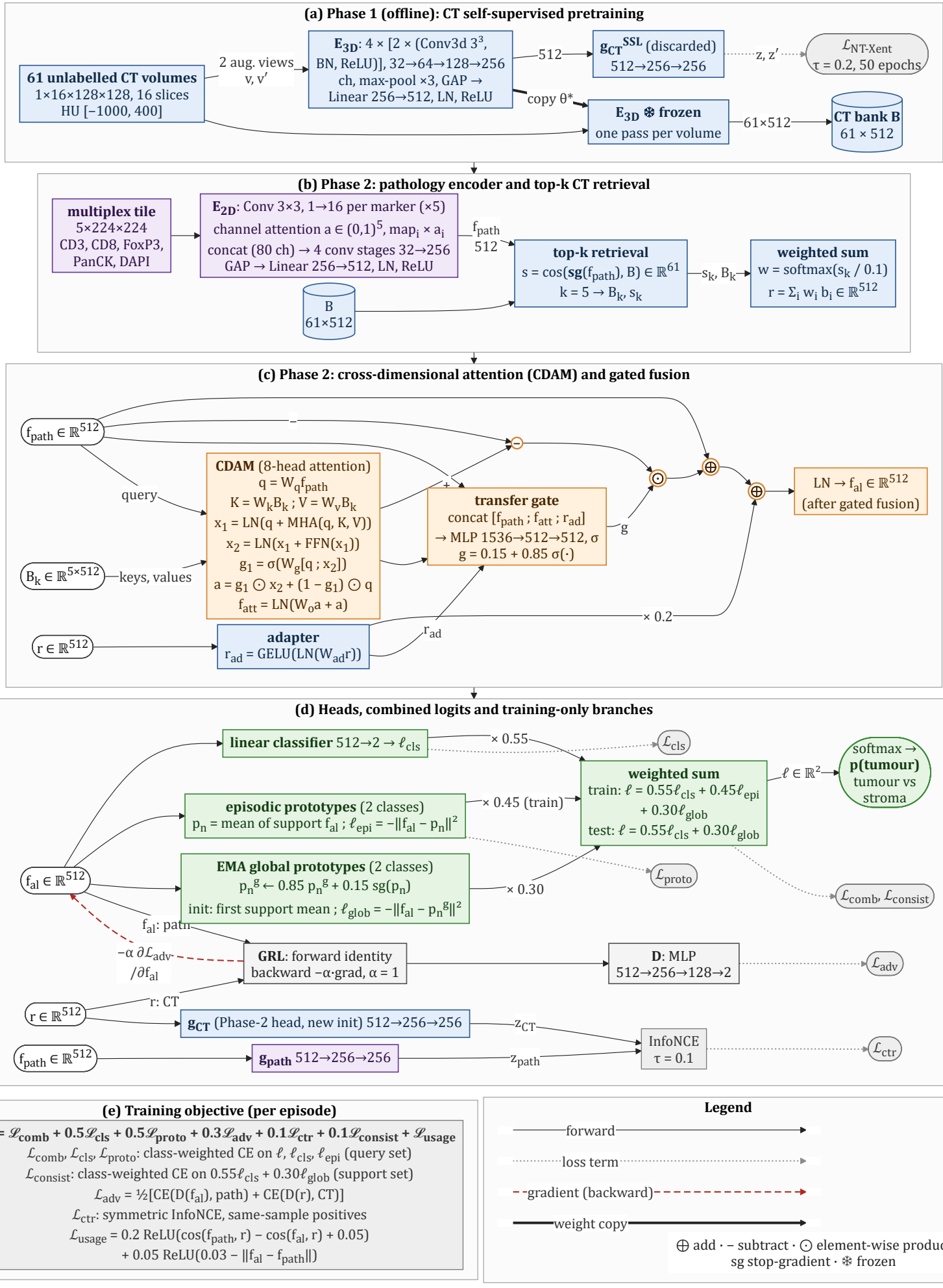

(a) Phase 1 (offline): CT self-supervised pretraining
61 unlabelled CT volumes
1×16×128×128, 16 slices
HU [−1000, 400]
2 aug. views
v, v′
E3D: 4 × [2 × (Conv3d 3³, BN, ReLU)], 32→64→128→256 ch, max-pool ×3, GAP → Linear 256→512, LN, ReLU
512
gCT SSL (discarded)
512→256→256
z, z′
ℒNT-Xent
τ = 0.2, 50 epochs
copy θ*
E3D ❄ frozen
one pass per volume
61×512
CT bank B
61 × 512
(b) Phase 2: pathology encoder and top-k CT retrieval
multiplex tile
5×224×224
CD3, CD8, FoxP3, PanCK, DAPI
E2D: Conv 3×3, 1→16 per marker (×5)
channel attention a ∈ (0,1)⁵, mapi × ai
concat (80 ch) → 4 conv stages 32→256
GAP → Linear 256→512, LN, ReLU
fpath
512
B
61×512
top-k retrieval
s = cos(sg(fpath), B) ∈ ℝ⁶¹
k = 5 → Bk, sk
sk, Bk
weighted sum
w = softmax(sk / 0.1)
r = Σi wi bi ∈ ℝ⁵¹²
(c) Phase 2: cross-dimensional attention (CDAM) and gated fusion
fpath ∈ ℝ⁵¹²
query
Bk ∈ ℝ⁵ˣ⁵¹²
keys, values
CDAM (8-head attention)
q = Wqfpath
K = WkBk ; V = WvBk
x1 = LN(q + MHA(q, K, V))
x2 = LN(x1 + FFN(x1))
g1 = σ(Wg[q ; x2])
a = g1 ⊙ x2 + (1 − g1) ⊙ q
fatt = LN(Woa + a)
transfer gate
concat [fpath ; fatt ; rad]
→ MLP 1536→512→512, σ
g = 0.15 + 0.85 σ(·)
g
r ∈ ℝ⁵¹²
adapter
rad = GELU(LN(Wadr))
rad
× 0.2
LN → fal ∈ ℝ⁵¹²
(after gated fusion)
(d) Heads, combined logits and training-only branches
linear classifier 512→2 → ℓcls
× 0.55
ℒcls
episodic prototypes (2 classes)
pn = mean of support fal ; ℓepi = −‖fal − pn‖²
× 0.45 (train)
EMA global prototypes (2 classes)
pn^g ← 0.85 pn^g + 0.15 sg(pn)
init: first support mean ; ℓglob = −‖fal − pn^g‖²
× 0.30
ℒproto
weighted sum
train: ℓ = 0.55ℓcls + 0.45ℓepi + 0.30ℓglob
test: ℓ = 0.55ℓcls + 0.30ℓglob
ℓ ∈ ℝ²
softmax → p(tumour) tumour vs stroma
ℒcomb, ℒconsist
fal ∈ ℝ⁵¹²
fal: path
−α ∂ℒadv /∂fal
GRL: forward identity backward −α·grad, α = 1
D: MLP 512→256→128→2
ℒadv
r ∈ ℝ⁵¹²
r: CT
gCT (Phase-2 head, new init) 512→256→256
zCT
fpath ∈ ℝ⁵¹²
gpath 512→256→256
zpath
InfoNCE τ = 0.1
ℒctr
(e) Training objective (per episode)
ℒ = ℒcomb + 0.5ℒcls + 0.5ℒproto + 0.3ℒadv + 0.1ℒctr + 0.1ℒconsist + ℒusage
ℒcomb, ℒcls, ℒproto: class-weighted CE on ℓ, ℓcls, ℓepi (query set)
ℒconsist: class-weighted CE on 0.55ℓcls + 0.30ℓglob (support set)
ℒadv = ½[CE(D(fal), path) + CE(D(r), CT)]
ℒctr: symmetric InfoNCE, same-sample positives
ℒusage = 0.2 ReLU(cos(fpath, r) − cos(fal, r) + 0.05) + 0.05 ReLU(0.03 − ‖fal − fpath‖)
Legend
forward
loss term
gradient (backward)
weight copy
⊕ add · − subtract · ⊙ element-wise product
sg stop-gradient · ❄ frozen